Theorem (Pythagorean theorem)**.** *In a right triangle with legs of lengths a and b and hypotenuse of length c,*

$$a^2 + b^2 = c^2.$$

Proof. Let ABC be a triangle with a right angle at C, and let D be the foot of the altitude from C to AB. By similarity of triangles,

$$AC^2 = AB \cdot AD, \qquad BC^2 = AB \cdot DB.$$

Adding these identities gives

$$AC^2 + BC^2 = AB(AD + DB) = AB^2.$$

Taking $a = AC$, $b = BC$, and $c = AB$ proves the result. □